Sachin Kumar, Ph.D.

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Executive Summary:

Senior Engineering Leader with over 15 years of R&D and product/process development experience across lithium-ion, lead-acid battery, and carbon capture technologies. Proven track record of scaling clean energy concepts from lab prototyping to high-volume manufacturing. Expert in electrochemistry, cell assembly, and leading cross-functional engineering teams.

Core Technical Competencies:

Electrochemical Engineering & Energy Storage, Process Development & Materials Scale-Up, Analytical Metrology & Characterization, Simulation & Data Analysis

Experience:

Independent Consultant | Scotts Valley, CA
Aug 2026 - Current

- Electrochemical & Clean Energy Advisory Services – Advised deep tech Fuel-Cell startup client under strict NDA on electrochemical testing of their novel electrode design and materials, accelerating their prototype timeline.
- Critical Minerals & Supply Chain Advisory Services – Conducted technical due diligence and supply chain risk assessment for a private equity firm evaluating domestic Graphite manufacturing project.

Prometheus Fuels | Santa Cruz, CA
Sr. Staff Process Engineer | July 2024 – July 2026

- Scale-up atmospheric CO_2 conversion technology into net-zero carbon fuels - Successfully demonstrating 1, 5 and **50 cell stacks for syngas production with Stoichiometric ratio of 2**.
- Directed R&D lab operations and instrument calibration for critical analytical systems.
- Lead cross-functional engineering teams to develop, coat, and analyze next-gen electrodes.
- Automated material handling by programming and integrating a 6-axis robotic arm.
- Provided operational and technical data pipelines directly to the Founder/CEO.

Tesla, Inc. | Fremont, CA
Sr. Staff Process Development Engineer | November 2022 – June 2024

- Scaled cell manufacturing for the 4680-battery program - Process successfully validated and scaled at GFTX for **Model Y and Cybertruck battery packs**.
- Accelerated 4680 cell wetting by scaling dry battery electrode manufacturing from the electrolyte fill side, achieving a **50% reduction in E-Fill time to match wet electrode performance**.
- Developed proprietary multi-stage electrolyte filling hardware for rapid cell wetting.
- Co-designed laser-structured electrodes to maximize high-power cell output.
- Partnered with the Cell Engineering Manager and materials teams on core process goals.

Microporous LLC | Piney Flats, TN
Technical and R&D Manager | May 2021 – November 2022

- Managed product development for specialized rubber and hybrid industrial battery separators.
- Accelerated commercialization of new formulations under harsh chemical conditions.
- Oversaw central R&D operations, standard testing procedures, and laboratory safety protocols.
- Represented the company on global technical committees to establish industry benchmarks.
- Reported to VP of Technology to align R&D initiatives with business goals.

Hollingsworth & Vose Company | West Groton, MA
Senior Scientist | July 2014 – April 2021

- Co-invented PowerAmp and Titan pasting papers for advanced automotive batteries, scaling product lines from initial concept into **$31.5M+ in combined annual revenue**.
- Managed cross-functional projects developing EFB separators for Start-Stop vehicles.
- Created electrochemical testing methods adopted globally as a standard by the BCI.
- Evaluated competitor patents and shaped IP strategies to mitigate commercial risks.

Earlier Career History:

- **Senior Research Scientist** | Shamrock Energy Corporation | 2012 – 2014
- **Postdoctoral Research Scientist** | University of Wisconsin | 2010 – 2012
- **Visiting Assistant Professor** | Miami University | 2008 – 2009
- **Research/Teaching Associate** | Miami University | 2001 – 2008
- **Executive Engineer** | Ballarpur Industries Ltd. (BILT), India | 1999 – 2001

Education & Credentials:

- **Ph.D., Analytical Chemistry** – Miami University, Oxford, OH (2008)
- **M.S., Paper Science and Engineering** – Miami University, Oxford, OH (2004)
- **B.E., Pulp and Paper Technology** – Indian Institute of Technology, Roorkee, India (1999)
- **Certifications**: Mini MBA – Miami University (2021), Minitab 40-Hour Training

Patents:

- **Patent (US 11,804,634 B2)**: Battery components comprising fibers (Issued Oct 31, 2023).
- **Patent (US 11,289,771 B2)**: Battery components comprising fibers (Issued Mar 29, 2022).
- **Patent (US 10,135,051 B2)**: Battery components comprising fibers (Issued Nov 20, 2018).
- **Patent (US 10,014,501 B2)**: Battery separators having a low apparent density (Issued July 3, 2018).
- **Application (WO 2019241338 A8)**: Pasting papers and capacitance layers for batteries comprising multiple fiber types and/or particles (Published Feb 6, 2020).
- **Application (US 20190181410 A1)**: Pasting papers and capacitance layers for batteries comprising multiple fiber types and/or particles (Published June 13, 2019).

Publications:

1. "Electroreduction of O_2 on uniform arrays of Pt nanoparticles", ***J. Electroanal. Chem.*** (2013).
2. "Monodisperse and highly active PtNi nanoparticles for O_2 reduction", ***Electrochem. Comm.*** (2009).
3. "Electrooxidation of CO on uniform arrays of Au nanoparticles- Effect of particle size and interparticle spacing", ***Langmuir*** (2009).
4. "Vertically aligned ZnO rods with multi-stage terrace structure and their enhanced solar cell efficiency", ***Crystal Growth and Design*** (2008).
5. "Seed-mediated growth of uniform gold nanoparticle arrays", ***J. of Physical Chemistry*** **C** (2007).
6. "Electrooxidation of carbon monoxide and methanol on platinum overlayer coated gold nanoparticles - Effects of film thickness", ***Langmuir*** (2007).
7. "Electroreduction of O_2 on uniform arrays of Pt and PtCo nanoparticles", ***Electrochem. Comm.*** (2006).
8. "Surface-enhanced Raman spectroscopy studies of 1,4-phenylene diisocyanide adsorption on Au and Pt-group transition metals", ***J. of Physical Chemistry*** **B** (2006).
9. "Electrooxidation of carbon monoxide on gold nanoparticle ensemble electrodes: Effects of particle coverage", ***J. of Physical Chemistry*** **B** (2005).
10. "An investigation of nanostructured vanadia/titania catalysts for the oxidation of monochlorobenzene", ***Catalysis Today*** (2003).